# Developing a Rule-Based Stemmer for the Bangla Language Based on the Snowball Stemmer

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***Abstract*—Stemming is a fundamental preprocessing step in Natural Language Processing (NLP) that reduces inflected word forms to their common roots. Bangla is a morphologically rich language with approximately 300 million speakers. But it is still underserved in this field. This thesis presents a rule-based Bangla stemmer developed using the Snowball stemming framework, a widely adopted codebase for building and deploying stemming algorithms. The proposed stemmer encodes 98 suffix-stripping rules in total, implemented in the Snowball language using Unicode character definitions following the Avro Phonetic transliteration scheme. The stemmer is evaluated on a manually constructed dataset of inflected word–root pairs.**

***Index Terms*—Bangla stemmer, Snowball framework, morphological analysis, suffix stripping, rule-based stemming**

## I. Introduction

Stemming is a critical preprocessing step in many NLP applications including information retrieval, sentiment analysis, text classification, spell checking etc. While highly effective stemmers — such as the Porter Stemmer [1] and its successor, Snowball — exist for English and many European languages, Bangla (Bengali) has received comparatively limited attention despite being the seventh most spoken language in the world.

Bangla is a morphologically rich Indo-Aryan language. Its verb system alone encodes tense, aspect, person, formality, and mood through a complex formation of suffixes. Nouns change forms for number, definiteness, and other cases. Vowels come in two different ways — independent vowel letters and dependent vowel signs (*matras*). All these make Bangla stemming significantly more complex than stemming for European languages.

Existing Bangla stemming efforts [3]–[5] have been implemented as standalone programs or packages without integration into a standardized, reusable framework. The Snowball framework, developed by Martin Porter [2], provides a platform-independent language for writing stemming algorithms that can be compiled in C, Java, and Python. No published Snowball implementation for Bangla exists to date.

This work addresses that gap by developing Snowball-based Bangla stemmer, contributing to a reusable, ex- tensible, and openly available implementation.

## II. Main Results

### A. Stemmer Design

The proposed stemmer is implemented in the Snowball language. Intuitions for suffix-stripping approach are adapted observing the structures of Hindi Stemmer inside Snowball Stemmer as Hindi is another Indo-Aryan language and very native to Bangla. The Hindi Stemmer is implemented based on the approaches from [7].

**Unicode character definitions:** All Bangla Unicode codepoints (U+0980–U+09FF) are assigned human-readable names following the Avro Phonetic Keyboard 3.1 convention [6]. Two categories are defined for vowels — independent vowel letters (e.g., `a` for *aa* vowel, U+0986) and dependent vowel signs or *matras* (e.g., `_a` for *aa-kar*, U+09BE). This mirrors the dual-form structure of Bangla script.

**Groupings and guard routines:** A `consonant` grouping enumerates all 39 Bangla consonant codepoints. A `CONSONANT` backward-mode routine uses this grouping as a guard condition. While stemming the suffixes, sometimes it is necessary to check the immediate letter before the suffix, and to add condition on the letter to be consonant or not. This is basically consonant guarding which ensures that matra-initial suffix rules apply only when the character preceding the suffix is a consonant. This prevents incorrect stripping from vowel sequences that are part of the stem.

**Suffix rules:** The `stem` routine applies a longest-match backward scan across 98 suffix rules organized into two sets. First set covers 39 suffixes for nouns and adjectives (e.g., animate plural marker *-ra*, *-er* for genitive case). Second set covers 59 suffixes for verbs (e.g., verb future suffix *-ben*, *-ano* for causative verbal noun). Both sets of suffixes are protected by a CONSONANT guard where needed. Verb paradigms covered simple present (*-y*) , past continuous (*-chhil*), present continuous (*-chh*), present perfect (*-eChh*), simple past (-*l*), future (*-b*), habitual past/conditional (*-t*), imperative (*-n*), causative (*-ao*), optative (*-uk*), non-finite (*-ar*). A good number of suffix suggestions are taken from [8].

### B. Evaluation

The stemmer is evaluated on a manually constructed dataset. News articles from contemporary Bangla journalism (Prothom Alo) were collected. Inflected words were extracted and then verified manually to produce word–root pairs. The evaluation metrics used are:

$$\text{Accuracy} = \frac{\text{Correctly Stemmed Words}}{\text{Total Candidate Words}} \tag{1}$$

TABLE I
STEMMER PERFORMANCE ON DATASET

| Metric | Value |
|---|---|
| Total candidate words | 166 |
| Correctly stemmed | 144 |
| Over-stemmed | 8 |
| Under-stemmed | 14 |
| Accuracy | 86.7% |

The results confirm that the suffix rules handle the most frequent inflectional categories effectively. The primary sources of errors are (1) over-stemming of words where a legitimate suffix sequence matches a non-suffix character sequence within the stem, and (2) under-stemming of inflectional forms which are not covered by the current rule set.

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